



salesforce

CUSTOMER SUPPORT TICKET PRIORITY PREDICTION AND AUTOMATED ASSIGNMENT SYSTEM USING AGENTFORCE



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1. INTRODUCTION

1.1 Project Overview

The Customer Support Ticket Priority Prediction and Automated Assignment System uses Salesforce and Agentforce to analyze customer support ticket descriptions, automatically determine priority as High, Medium, or Low, and trigger backend automation for appropriate support handling.

The solution uses the Support Ticket Intelligence custom object, an Auto-Launched Flow, and an Agentforce subagent named Support Ticket Priority Analysis. High-priority tickets create an urgent handling task and are assigned to a senior support agent.

1.2 Purpose

- Automatically classify support tickets as High, Medium, or Low priority.
- Retrieve the latest support ticket associated with a customer Account.
- Reduce manual prioritization and assignment effort.
- Create an urgent handling task for High-priority tickets.
- Use Agentforce to analyze ticket details and trigger backend automation.
- Improve ticket response and resolution efficiency.

2. IDEATION PHASE

2.1 Problem Statement

User	Need	Problem
Support Agent	Focus on critical tickets first	Tickets are not always prioritized automatically; prioritization can be manual.
Customer	Get urgent issues resolved quickly	Critical issues may wait behind lower-priority tickets.
Support Manager	Assign tickets to the right support level	Assignment may require manual decisions and urgency may not be consistently analyzed.
Support Agent	Identify urgent issues from descriptions	Important keywords can be missed and descriptions contain

		different urgency signals.
Support Manager	Reduce SLA-related risk	Older unresolved tickets may remain unnoticed and need additional checking.

2.2 Empathy Map – Support Agent

- Think & Feel: Wants critical issues identified quickly so urgent customer problems receive attention.
- Hear: Receives customer issue descriptions and support requests.
- See: Ticket status, issue type, description, priority level, assignment, and SLA risk.
- Say & Do: Reviews tickets, handles assigned cases, and focuses on urgent support issues.
- Pain: Manual prioritization, missed urgency signals, and delays in handling critical tickets.
- Gain: Automatic priority classification, clearer assignment, urgent-task creation, and reduced manual effort.

2.3 Brainstorming & Idea Prioritization

- Manual ticket registers
- Spreadsheet-based ticket tracking
- Standalone support portal
- Salesforce CRM with Agentforce

The documented final selection is Salesforce CRM with Agentforce because it provides automation through Salesforce Flow, AI-assisted ticket analysis through Agentforce, automatic priority classification, automatic urgent-task creation, and reduced manual support effort.

3. REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey

Ticket Creation → Account Identification → Ticket Retrieval → Description Analysis → Priority Classification → Agent Assignment → Urgent Task Creation → Support Handling → Resolution

3.2 Functional Requirements

ID	Requirement
FR-1	Support ticket creation and management
FR-2	Account and Contact association
FR-3	Ticket description analysis
FR-4	Automatic priority classification
FR-5	High-priority task creation
FR-6	Support agent assignment
FR-7	SLA breach risk checking
FR-8	Agentforce conversational analysis

3.3 Non-Functional Requirements

Requirement	Description
Usability	Clear Agentforce interaction and understandable priority messages.
Reliability	Consistent priority classification based on configured conditions.
Performance	Quick retrieval and analysis of the latest support ticket.
Security	Controlled Salesforce access to customer and ticket information.
Maintainability	Flow and Agentforce configuration can be updated.
Scalability	Supports increasing support-ticket volumes through Salesforce automation.

3.4 Data Flow

Customer / Support Request → Account → Latest Support Ticket → Ticket Description → Analyze Description → Priority Level → High/Medium/Low Decision → Agent Assignment → High-Priority Task (if applicable) → Final Response

Agentforce: User provides Account Name → Support Ticket Priority Analysis → Auto-Launched Flow → Ticket Analysis → Priority and Assignment Output → Response to User.

3.5 Technology Stack

Layer	Technology
Platform	Salesforce
Database	Support_Ticket_Intelligence__c with Account, Contact, User and Task records
Logic	Auto-Launched Flow
AI	Agentforce
Agent Configuration	Support Ticket Priority Analysis subagent
Automation	Flow Decision, Assignment and Create Records
Output	Agentforce conversation response

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

Problem	Solution	Benefit
Manual ticket prioritization	Flow analyzes description and sets priority	Faster identification of urgent cases
Urgent issues may be missed	High-priority keyword decision	Critical cases receive attention
Manual task creation	Flow creates Task for High priority	Reduced manual effort
Manual support assignment	Flow sets assigned support level	More consistent handling
SLA risk can be overlooked	Optional SLA Risk decision	Better risk awareness
Need for conversational access	Agentforce subagent	AI-assisted interaction

4.2 Proposed Solution

Parameter	Description
Problem	Manual ticket prioritization and assignment causing delays in handling critical customer support issues.
Solution	Salesforce-based Customer Support

	Ticket Priority Prediction and Automated Assignment System.
Innovation	Agentforce AI + Auto-Launched Flow for analyzing ticket descriptions, automatically classifying priority, assigning support level, and triggering urgent-task automation.
Impact	Faster identification of critical tickets, reduced manual effort, improved response time, and better support-team productivity.
Business Model	Customer support operations involving customers, support agents, support managers, and AI-assisted ticket handling.
Scalability	Supports increasing support-ticket volumes through Salesforce automation, with scope for additional assignment rules, SLA monitoring, and priority indicators.

4.3 Solution Architecture

Users / Support Team → Agentforce → Support Ticket Priority Analysis → Auto-Launched Flow → Account & Ticket Data → Description Analysis → Priority Decision → Task / Assignment → Final Message → Agentforce Response

5. SYSTEM IMPLEMENTATION

5.1 Salesforce Data Model

The documented implementation uses the Support Ticket Intelligence custom object. Key Salesforce records are Account, Contact, User, and Task.

- Support Ticket Intelligence: stores support-ticket information used by the automation.
- Account: identifies the customer account.
- Contact: supports the account/contact association.
- User: represents Salesforce users involved in support handling.
- Task: records urgent handling work for High-priority cases.

5.2 Auto-Launched Flow

Flow name in the source appendix: Support_Ticket_Intelligence. The flow is documented as an Auto-Launched Flow.

Step	Flow Element	Purpose
1	Get Account	Find the Account using the provided Account Name.
2	Store Account Id	Store the Account Id in varAccountId.
3	Get Ticket	Retrieve the latest Support Ticket Intelligence record for the Account.
4	Store Ticket Id	Store the ticket Id in varTicketId.
5	Analyze Description	Check ticket description and select High, Medium, or Low path.
6	Set Priority Level	Set varPriorityLevel to High, Medium, or Low.
7	High Priority Check	Check whether varPriorityLevel equals High.
8	Create Task	Create an Urgent Ticket Handling task for High-priority tickets.
9	Set Assigned Agent	Set assigned support level as Senior Support Agent.
10	Optional SLA Risk	Check whether an older unresolved ticket may create SLA risk.
11	Final Message	Set the final response message for Agentforce.
12	Save Flow	Save and activate the configured flow.

5.3 Priority Classification Logic

Priority	Keywords	Result
High	urgent; not working; failure	Set High; create urgent task; assign senior support level.
Medium	issue; slow; delay	Set Medium.
Low	None of the High or Medium keywords	Set Low.

5.4 High-Priority Task

- Object: Task
- Subject: Urgent Ticket Handling
- WhatId: varTicketId
- Priority: High
- Status: Not Started

5.5 Final Response Messages

Priority	Final Message
High	High priority ticket detected. Assigned to senior agent.
Medium	Ticket marked as medium priority. Will be handled shortly.
Low	Tickets are low priority and queued for processing.

6. AGENTFORCE CONFIGURATION

6.1 Agentforce Subagent

Field	Configured Value
Name	Support Ticket Priority Analysis
API Name	Support_Ticket_Priority_Analysis
Classification Description	This topic analyzes customer support tickets to determine priority based on

	issue description and urgency. It handles user requests related to ticket status, issue severity, and support prioritization.
Scope	Analyze support ticket descriptions, determine High/Medium/Low priority, and trigger backend automation for task assignment.

6.2 Subagent Instructions

- Ask the user to provide the Account Name if not already given.
- Retrieve the latest support ticket associated with the Account.
- Read and analyze the ticket description.
- Determine priority using the configured keywords.
- Set the priority level accordingly.
- For High Priority, trigger the Auto-Launched Flow and ensure a task is created for immediate handling.
- Assign the ticket to the appropriate support level, such as a senior agent for High priority.
- Return a clear response indicating urgent handling, short-term handling, or queueing.
- Do not update records directly or send emails; use Flow for actions.

6.3 Agentforce Action

The documented Agentforce action is a Flow-based Support Ticket Intelligence action. It analyzes customer support tickets using the provided Account Name, determines priority from the issue description, and triggers automation for High-priority cases.

Variable	Role	Setting
varAccountName	Text input used to find the customer Account	Required input; collect from user
varAccountId	Account identifier returned by the Flow	Show in conversation; Text output
varActionMessage	Final message indicating priority and actions	Show in conversation; Text output
varAssignedTo	Assigned support	Show in conversation;

	agent/team	Text output
varPriorityLevel	Calculated High / Medium / Low priority	Show in conversation; Text output
varTicketId	Unique identifier of analyzed support ticket	Show in conversation; Text output

7. END-TO-END WORKFLOW

1. User provides the customer Account Name through Agentforce.
2. Agentforce invokes the Support Ticket Priority Analysis action.
3. The Auto-Launched Flow retrieves the Account.
4. The Flow retrieves the latest support ticket for that Account.
5. The ticket description is checked against the configured priority keywords.
6. The Flow sets the priority as High, Medium, or Low.
7. If High priority, the Flow creates an Urgent Ticket Handling task.
8. The Flow sets the assigned support level to Senior Support Agent for the High-priority path.
9. The Flow prepares the final response message.
10. Agentforce displays the result to the user.

8. FUNCTIONAL AND PERFORMANCE TESTING

The project documentation states that testing was performed in Salesforce and screenshots were captured to validate configured features and their functionality and performance.

Test Area	Expected Result
Ticket description analysis	Description is checked against configured High, Medium, and Low keywords.
High-priority classification	Priority is High and an Urgent Ticket Handling task is created.
Medium-priority classification	Priority is Medium and the medium-priority response is returned.
Low-priority classification	Priority is Low and the low-priority response is returned.
Agent assignment	High-priority cases are assigned to the Senior Support Agent support level.

Milestone 2: Field & Relationship

Developer Edition

Search Setup

Setup Home Object Manager

SETUP > OBJECT MANAGER

Support Ticket Intelligence

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

14 Items, Sorted by Field Label

Quick Find

New Deleted Fields Field Dependencies Set History Tracking

Created By	CreatedById	Lookup(User)	
Created Date	Created_Date__c	Date	
Customer	Customer__c	Lookup(Account)	✓
Description	Description__c	Long Text Area(32768)	
Issue Type	Issue_Type__c	Picklist	
Last Modified By	LastModifiedById	Lookup(User)	
Owner	OwnerId	Lookup(User,Group)	✓

Objects Creation

SkillWallet x Develop: x salesforce x Support T x Agentfor: x Support T x Agentfor: x Support T x gmail - S x Welcome x

https://orgfarm-90b0d1965c-dev-ed.develop.my.salesforce-setup.com/lightning/setup/ObjectManager/01lgK000007gyRx/Details/view

Developer Edition

Search Setup

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Details

Edit Delete

Description

API Name

Support_Ticket_Intelligence__c

Custom

✓

Singular Label

Support Ticket Intelligence

Plural Label

Support Ticket Intelligence

Enable Reports

✓

Track Activities

Track Field History

✓

Deployment Status

Deployed

Help Settings

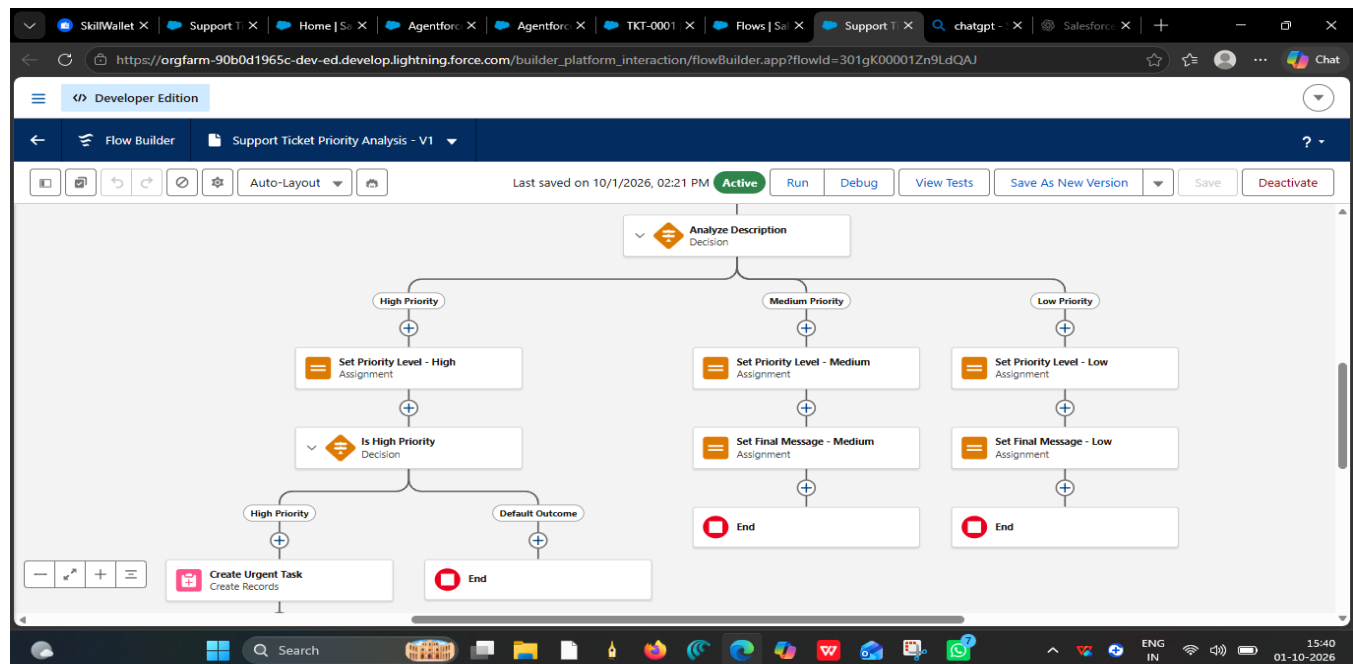
Standard salesforce.com Help Window

Milestone 3: Creating a tab for Support Ticket Intelligence Object

The screenshot shows the Salesforce Setup interface in Developer Edition. The left sidebar contains the 'App Launcher' search bar and a list of items, with 'Support Ticket Intelligence' selected under the 'Items' section. The main content area displays the 'Tabs' configuration page for the 'Support Ticket Intelligence' object. It includes sections for 'Object Tabs', 'Web Tabs', 'Visualforce Tabs', and 'Lightning Component Tabs'. The 'Object Tabs' section shows a table with one tab defined: 'Support Ticket Intelligence' with a 'Laptop' tab style.

Label	Tab Style	Description
Support Ticket Intelligence	Laptop	

Flow Builder (Support Ticket Priority Analysis)



VarAccountName ABC Technologies:(Run)

Developer Edition

Flow BuilderSupport Ticket Priority Analysis - V1

Last saved on 10/1/2026, 02:21 PMActiveRunDebugView TestsSave As New VersionSaveDeactivate

Debug

Setup

Input Variables

Enter values for the flow's input variables. For each value left blank, the flow starts with the variable's default value. You can't enter values for collection variables or Apex-defined variables.

varAccountName

ABC Technologies

Select Debug Options

☒ Debug wait element behavior

☐ Run automation as another user

Reset Debug SettingsRun

Autolaunched Flow

Start

+

Get Account

Get Records

+

Store Account ID

Assignment

+

Get Ticket

Get Records

Support Ticket Priority Analysis Flow Completed:

Developer Edition

Flow BuilderSupport Ticket Priority Analysis - V1

Debug Run: Support Ticket Priority Analysis 3/10/2026, 11:40 amCompletedEdit FlowConvert to TestSave As New VersionSaveDeactivate

Debug

SetupDetails

Expand AllBasic Debug Log

Search this list...

How the Interview Started

Details

Get Records: Get Account

One or more Account records were retrieved.

Details

Convert to Test

Autolaunched Flow

Start

+

Get Account

Get Records

+

Store Account ID

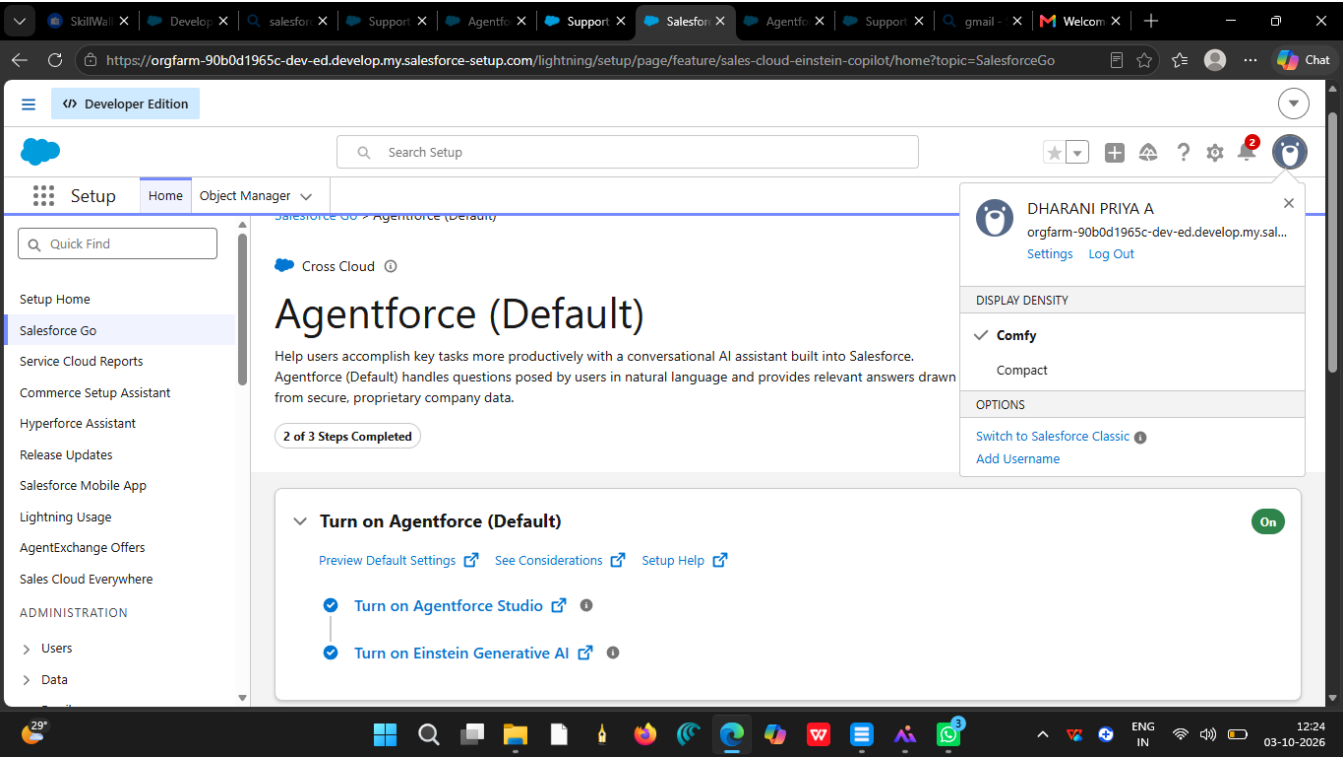
Assignment

+

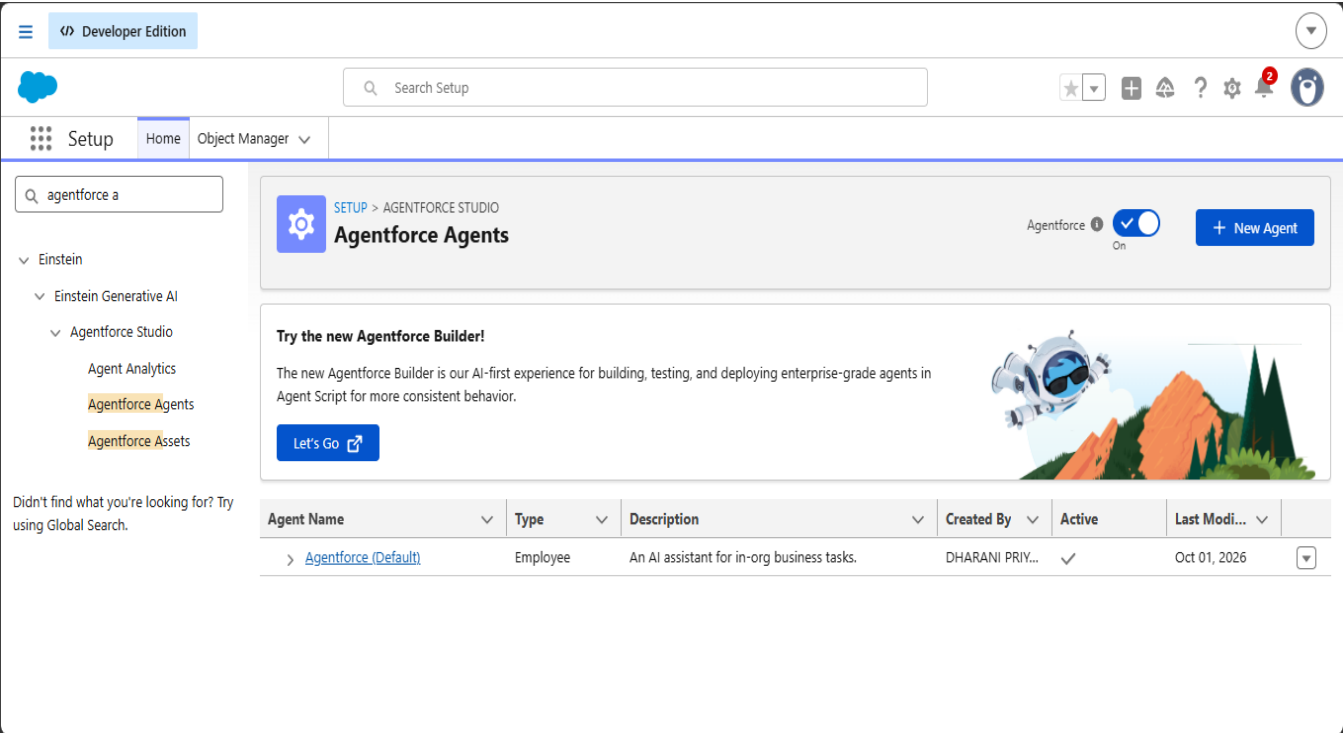
Get Ticket

Get Records

Org Setup for AgentForce:



Building the Agent:



Adding Sub Action:

The screenshot shows the Salesforce Agentforce Builder interface. The top navigation bar includes 'Developer Edition', 'Agentforce Builder', 'Agentforce (Default)', and 'Version 1 (Active)'. The main content area is titled 'View Action' and displays the configuration for the 'Support Ticket Priority Analysis' action. The configuration includes the Agent Action API Name, Last Modified date, and Assigned to Active Agent status. The Agent Action Configuration section describes the action's purpose: to analyze customer support tickets based on the provided account name, determine ticket priority using issue description, and trigger automation to assign tasks for high-priority cases. The Agent Action Label is 'Support Ticket Priority Analysis'. The Agent Action Description is: 'Analyzes customer support tickets based on the provided account name, determines ticket priority using issue description, and triggers automation to assign tasks for high-priority cases.' The 'Require user confirmation' checkbox is unchecked. A 'Conversation Preview' window on the right shows a chat interface with a cartoon robot character and the text: 'Let's chat! Hi! I'm Agentforce, an AI assistant. I can do things like search your Salesforce data or the internet, edit your Salesforce records, and draft or revise emails and fields.... Describe your task or ask a question...'

Verify the Agent:

The screenshot shows the Salesforce Agentforce Studio interface. The top navigation bar includes 'Developer Edition', 'Agentforce Studio', 'Support Ticket Intelligence', and 'Support Ticket Priority Analysis'. The main content area is titled 'Support Ticket Priority Analysis' and displays the details of the action. The 'Information' section includes the Flow Label, Description, Associated Record, Created By, Last Modified, Category, and Overridden By. The 'Details' section includes the API Name, Flow Type, Segment, Created Date, Last Modified Date, Subcategory, and Overrides. A 'Recently Viewed' list on the left shows the 'TKT-0001' item. A chat preview window on the right shows a chat interface with a cartoon robot character and the text: 'Analyze support ticket for ABC Technologies. The support ticket for ABC Technologies has been analyzed and assigned for immediate handling. 001gK00001ZmPAJQAN. High priority ticket detected. Assigned to senior agent. Senior Support Agent. true. a00gK00001O6FE9QAN. Describe your task or ask a question...'

9. ADVANTAGES

- Automatic Ticket Prioritization – Classifies support tickets as High, Medium, or Low based on ticket description.
- Faster Identification of Critical Issues – Helps support agents focus on urgent tickets first.
- Reduced Manual Effort – Automates ticket analysis, priority setting, task creation, and assignment.
- Intelligent Ticket Handling – Agentforce analyzes ticket details and initiates the appropriate backend automation.
- Automatic Urgent Task Creation – High-priority tickets automatically create an Urgent Ticket Handling task.
- Improved Support Assignment – Assigns High-priority tickets to the appropriate support level, including a senior support agent.
- Conversational Support – Support users can interact with Agentforce to analyze ticket information and receive the outcome.
- Better Response Time – Automation helps reduce delays in handling critical customer issues.
- SLA Risk Awareness – The solution can check older unresolved tickets for potential SLA breach risk.
- Scalable Support Operations – Salesforce automation can support increasing volumes of support tickets.
- Improved Team Productivity – Reduces repetitive work and allows support teams to concentrate on ticket resolution.

10. LIMITATIONS / DISADVANTAGES

- Keyword-Based Priority Logic – Priority classification depends on configured keywords such as urgent, not working, failure, issue, slow, and delay.
- Limited Understanding of Context – The current implementation is based on configured conditions and may not capture every possible urgency signal.
- Dependence on Accurate Ticket Data – Correct Account, Contact, and ticket information is required for successful analysis.
- Flow Configuration Dependency – Changes to ticket handling require maintenance of the Auto-Launched Flow and its logic.

- Agentforce Configuration Dependency – Agentforce responses and actions depend on properly configured subagent instructions and Flow actions.
- Limited Scope of Agentforce – The documented subagent focuses on ticket analysis, priority determination, and backend automation; unrelated requests such as billing, subscription management, or account updates are outside its scope.
- Limited SLA Monitoring – The current implementation includes an optional SLA-risk check rather than a comprehensive SLA monitoring and escalation solution.
- Assignment Logic May Need Expansion – Additional support teams or more sophisticated assignment rules would require further configuration.
- Maintenance Required – Priority keywords, Flow variables, Agentforce outputs, and assignment logic may need to be updated as support requirements evolve.
- Limited Analytics in Current Scope – Advanced analytics for priority trends and resolution performance are identified as future scope.

11. FUTURE SCOPE

- Add more priority indicators and business rules.
- Extend Agentforce outputs with additional ticket information.
- Add more sophisticated SLA monitoring and escalation.
- Support additional support teams or assignment rules.
- Introduce analytics for priority trends and resolution performance.
- Add more variables and configure them in the agent for richer outputs.

Team Members

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12. CONCLUSION

The Customer Support Ticket Priority Prediction and Automated Assignment System demonstrates how Salesforce Flow and Agentforce can work together to reduce manual support-ticket handling. The solution retrieves the latest ticket for an Account, analyzes its description, classifies priority, creates an urgent task for High-priority cases, assigns a support level, and returns a clear conversational response.

The documented solution is designed to support faster ticket handling, reduced manual effort, intelligent prioritization, and better team productivity.

13. APPENDIX

Item	Documented Configuration
Custom Object	Support Ticket Intelligence
Key Salesforce Records	Account, Contact, User, Task
Flow	Support_Ticket_Intelligence - Auto-Launched Flow
Agentforce Subagent	Support Ticket Priority Analysis
Agentforce Action	Flow-based Support Ticket Intelligence action

DEMOLINK:https://drive.google.com/file/d/1NRzo2ksGw5TUq_hhNYDritH4sFVSiMbQ/view?usp=drivesdk

— THANK YOU —